2025 IEEE 14th International Conference on Communication Systems and Network Technologies (CSNT) | 979-8-3315-3193-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/CSNT64827.2025.10967952

Deep Learning Enabled Secure IoT Module for Smart Agriculture in Diverse Environmental Conditions

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***Abstract*— The Internet of Things (IoT) stands as a revolutionary technology which impacts smart agriculture and different sectors beyond it. The research describes an intelligent IoT-enabled monitoring system which uses sensors to bring improved agricultural performance through environmental real-time observations combined with artificial intelligence decision systems. The proposed system completes its operation by executing sensor data acquisition followed by deep learning-based analysis and secure data transmission procedures. The initial stage of IoT-enabled operation utilizes environment-sensing devices which collect essential parameters for precision farming. A Recurrent Neural Network (RNN) processing system applies classification methods to collected data to generate useful insights. Secure cloud data transmission takes place through an encryption system which combines Substitution-Caesar cipher with Improved Elliptic Curve Cryptography (IECC). The system demonstrates 99.64% accuracy, and the proposed encryption method proves efficiency through its average execution times of encryption at 0.3904 ms and decryption at 0.1102 ms which exceeds current models in speed and security performance.**

Keywords—Internet of Things (IoT), Smart Agriculture, Recurrent Neural Network (RNN), Data Security

I. INTRODUCTION

Agriculture is widely recognized as a crucial industry for the production of food in the majority of countries. Nevertheless, it must confront a multitude of obstacles to enhance its sustainability[1]. The prudent exploitation of natural resources, the destruction of the environment, and climate change pose significant difficulties. Challenges in agriculture are encountered. Therefore, it is crucial to implement intelligent agricultural techniques to overcome the obstacles [2]. The application of IoT in agriculture or farming aims to deliver accurate and valuable data to farmers, including information on weather conditions, soil properties, fertilizer requirements, and seasonal recommendations for planting [3]. This data is crucial for enhancing production in the agricultural sector. IoT sensor devices generate substantial volumes of data on a daily basis, resulting in the creation of big data characterized by its volume, velocity, and variance [4]. Intelligently processing and analyzing this large dataset facilitates the development of smart applications [5] [26]. Nevertheless, without robust security measures, the potential risks and failures of the IoT may surpass its advantages[6]. Security and privacy encompass various concerns, including confidentiality, authentication, validation, non-repudiation, and integration[7].

Several key factors, namely air quality, soil quality, moisture, humidity, and temperature, influence agriculture's productivity. Smart agriculture implements IoT systems to enhance farming operations and reduce the time and effort required by farmers by perceiving, recognizing, and overseeing these elements[8]. Fig. 1 shows the readings of different sensors used in agriculture.

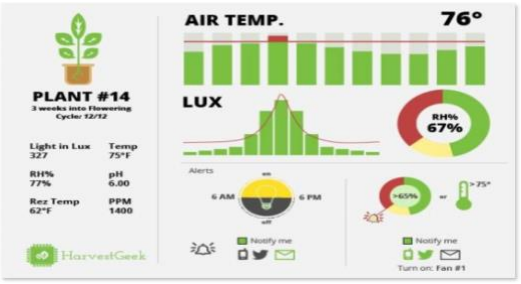


Fig. 1. Temperature and environmental sensors

The Internet of Things is a network of interconnected devices that can communicate and exchange data with each other[9]. The IoT is an innovative technology that enables the interconnection of many applications across different sectors using sensors and other devices[10]. Automated data collection minimizes the need for manual intervention and simplifies daily tasks[11][12]. Fig. 2 illustrates the functionalities of the IoT in smart agriculture and farming.

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DOI: 10.1109/CSNT.2025.16

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Fig.2. IoT in Smart Agriculture.

The aim of the study is to securely deliver processed sensor data to farmers. To achieve this, the data undergoes encryption before being sent to the cloud. Farmers can then remotely access this data from the cloud whenever needed.

This paper organizes the remaining sections as follows: Section 2 presents related works, Section 3 describes the proposed approach, Section 4 elaborates on the experimental findings, and Section 5 concludes the paper.

II. RELATED WORK

In recent years, there has been an increased focus on smart agriculture approaches, leading to much research in this field. Smart agricultural systems must adapt to climatic circumstances in order to achieve optimal crop yield. Appropriate Anticipating and cultivating crops can mitigate losses for farmers. In the same way, problems related to environmental devolution, such as the excessive extraction of groundwater and use of nitrogen fertilizers, will result in future challenges [13]. The literature discusses climate smart agriculture, which addresses the challenges posed by climate change and the efficient use of limited resources [14]. The assessment method of climate smart agriculture takes into account the multi-dimensional nature of the topic, including adaption, mitigation, productivity, uncertainties of climate impacts, as well as scale and temporal dependencies. This approach aims to effectively address the various concerns related to smart agriculture. In addition, the scaling of climate smart agriculture assists in the analysis of seed systems and rice varieties to enhance yields in comparison to standard farming practices [15].

This survey finds that existing research mainly concentrates on limited factors like humidity and temperature in smart agriculture systems. Only a few studies utilize machine learning for decision support. Our research introduces an integrated module incorporating secure monitoring systems with various IoT sensors, including UV, humidity, temperature, light intensity, and soil moisture sensors[16][17]. This novel approach offers farmers a comprehensive, secure smart model.

III. PROPOSED WORK

The suggested smart agricultural module has three distinct phases. In the initial stage, an IoT application employs several sensors, including UV, humidity, temperature, light intensity, and soil moisture sensors, to gather crucial environmental data. A deep learning-powered RNN algorithm analyzes and categorizes the gathered data during the second step. The third phase involves securely communicating the processed data to both the cloud and the end user. Using a secret key, the cloud and the end user can decrypt the data. Fig. 3 depicts the comprehensive workflow of the proposed IoT-based smart agriculture module.

The application utilizes wireless data transmission to remotely monitor the fields. We employ a microcontroller to establish connections with all the sensors and evaluate the gathered data using a one-dimensional recurrent neural network model to derive significant decisions. We encrypt the data using a hybrid model that combines the Substitution-Caesar cipher and improved Elliptical Curve Cryptography (IECC) before transmitting it to both the cloud and the end user. At their ends, a private key is used to decrypt the data. This novel method allows the user to retrieve historical data from the cloud. The cloud server displays the sensor data and an action message to the user based on sensor values. The user can receive real-time data directly from the microcontroller.

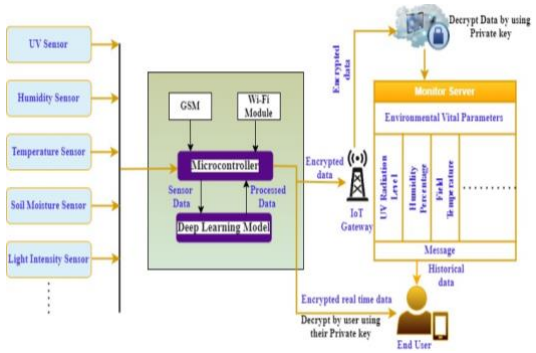


Fig. 3. Proposed Agriculture Module: A Comprehensive Perspective

A. *Sensors data Analysis using Recurrent Neural Network (RNN)*

For analysis of time series real-time data, recurrent neural networks (RNN) are the optimal choice[18][19]. RNN incorporates both current and previous state variables to make decisions. The reliability of the results produced by RNN is superior when compared to other models. The suggested smart agricultural model utilizes RNN to analyze the sensors data. A Recurrent Neural Network (RNN) is a specialized type of neural network that extends the structure of an artificial neural network by incorporating cyclic connections within its units. Unlike traditional machine learning models, which focus on direct input-to-output mapping, RNNs leverage past inputs to determine the current state and network parameters for each output. This temporal dependency makes RNNs particularly suitable for sequential data analysis.

The process of constructing a recurrent neural network for single-sensor time series data involves applying similar steps to all the data generated by the sensors. For a given time series dataset $X=(x_1, x_2, x_3, \ldots, x_n)$, the RNN processes the sequence by iteratively updating its internal states based on the input data, capturing temporal correlations effectively. Initially, the input data undergoes compression, resulting in vectors with defined dimensions. To encode the data into a vector representation, the system utilizes a recursive

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function, enabling it to handle varying input and output lengths effectively. The historical information of all vectors preceding the current state is represented as ht−1ht−1. The Recurrent Neural Network (RNN) computes the internal state by consolidating all prior states into a new vector, which serves as a compressed representation of the sequence history. This process facilitates the generation of the next symbol, represented mathematically as:

$$X_t = (x_1, x_2, x_3, x_4 \dots \dots \dots x_{t-1}) \quad (1)$$

The internal state is described as

$$h_t = tanh(WMX_t + RWh_{t-1} + b) \quad (2)$$

The output state is described as

$$\widehat{Y}_t = softmax(Uh_t) \quad (3)$$

Where WM is the input weight matrix, RW denotes the recurrent weight matrix, U denotes the hidden layer weight function, and b denotes the bias vector. Fig. 4 illustrates the architecture of a Recurrent Neural Network (RNN) with a single hidden layer, showcasing its operation across multiple time intervals.

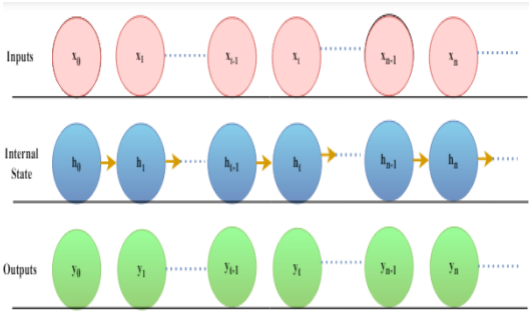


Fig.4. Single-layer Hidden Recurrent Neural Network Across Multiple Time Steps

The suggested approach utilizes the backpropagation technique to train the RNN module. The recurrent neural network receives the parameters in the form of timestamps, and the current and previous timestamps determine the output. To perform backpropagation, the error gradients must be computed with respect to the network parameters. This is achieved using the chain rule in differentiation, which is expressed as:

$$\frac{\partial h_t}{\partial U} = \frac{\partial \widehat{Y}_t}{\partial U} * \frac{\partial h_t}{\partial \widehat{Y}_t} = (\widehat{Y}_t - Y_t) \otimes h_t \quad (4)$$

The above mathematical function rely on the current timestamp values, such as h_t, Y_t and $\widehat{Y}_t$. The gradient of previous state is given as

$$\frac{\partial h_t}{\partial WM} = \sum_{j=0}^{t} \frac{\partial \widehat{Y}_t}{\partial H_t} \frac{\partial h_t}{\partial \widehat{Y}_t} \left(\prod_{i=j+1}^{t} \frac{\partial h_i}{\partial h_{i-1}} \right) \frac{\partial h_i}{\partial WM} \quad (5)$$

Equation 5 presents the chain rule using the inner product for the limit j. Gradients are added for each time stamp.

B. *Data Security*

After the RNN analyzes and categorizes the various sensor data, the microcontroller transmits the processed output sensor data to the cloud server and to the end user directly connected to it [20]. To transfer the processed data from the RNN's output, security is required. We propose two levels of encryption to ensure the privacy and security of the data during transfer. We first use the substitution Caesar cipher to encrypt the processed data, and subsequently, the IECC technique encrypts the ciphered text.

- *Substitution Caesar Cipher*

Initially, a basic substitution cipher encrypts the processed data obtained from the microcontroller. The replacement cipher, a technique in cryptography, encrypts plain text by substituting it with ciphertext [21][22]. The proposed approach employs a simple substitution cipher. The system autonomously replaces individual letters. You can illustrate simple substitution by arranging the alphabetic letters in a specific sequence to represent the substitution. This is classified as a substitution cipher. Complex manipulations can reverse, shift, or jumble a cipher alphabet, resulting in a mixed or de-ranged alphabet. Following the substitution cipher, the Caesar cipher further encrypts the resulting text. The Caesar cipher receives the output from the substitution cipher. We employ a substitution cipher that shifts each letter in the plain text by a specific number of alphabetic positions. For instance, we swap the letter A for the letter B, the letter B for the letter C, and so on. A key is employed in the Caesar cipher to shift the message. In this context, the key refers to a randomly selected number between 1 and 26. Equation (6) describes how this number is used to change the cipher alphabet. We obtain the encrypted data by applying the substitution and Caesar cipher, and then further encrypt it using the enhanced ECC technique before transmitting it to the cloud server and end user.

$$E(x) \equiv (x + K) \ (mod 26) \quad (6)$$

The variable K represents the shifting key for each letter, while x denotes the original plaintext. The decryption process can be formulated as illustrated in Equation (7), as follows:

$$D(x) \equiv (x - K) \ (mod 26) \quad (7)$$

- *Improved Elliptic Curve Cryptography (IECC)*

The IECC is a curve-based cryptographic technique that uses elliptic curves to obtain a unique base point through the manipulation of prime numbers. Moreover, the ECC method is more complex and challenging to execute, leading to a higher likelihood of implementation errors, hence reducing the security of the algorithm[23]. Thus, to improve security, we use the improved ECC (IECC) encryption method to encrypt the data. Traditional ECC generates two types of keys, namely public and private keys [24]. However, Improve ECC introduces a new key, known as the secret key, to enhance the system's security. The secret key is added to the encryption formula and subtracted from the decryption formula. Thus, the complexity of the two phases has increased. When the complexity of encryption and

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decryption is increased, it becomes extremely difficult to identify the original data. It autonomously improves the level of data security. Equation (8) demonstrates the mathematical representation of the enhanced ECC.

$$y^2 = x^3 + ax + b \quad (8)$$

Where a and b are integers.

The proposed system generates three distinct keys: a private key, a public key, and a secret key. The public key is derived from the private key and is utilized for encrypting data. This private key is employed in decrypting the data. The private key is then converted into a point on the elliptic curve. Meanwhile, the secret key is formed through a combination of the private key, the public key, and the global point of an elliptic curve.

Let G_s represents a global point on the curve. Generate a random number within the range of 0 to n-1 to create a private key, denoted as P_k. The public key P_u is generated according to Equation (9)

$$P_u = \prod(P_k, G_s) \quad (9)$$

The secret key S_k is created by the summation of P_u, P_k and G_s, according to the Equation (10):

$$S_k = \sum(P_u, P_k, G_s) \quad (10)$$

The obtained processed sensors data undergoes encryption by using the public key at the sender end following the generation of the secret key. The encrypted information contains two ciphertexts that are mathematically represented as Equations (11) and (12).

$$C_1 = (K * G_s) \quad (11)$$

$$C_2 = M + (K * P_u) + S_k \quad (12)$$

Where C_1 and C_2 represent the first and second ciphertexts, respectively. K is a randomly generated number that ranges from 1 to n-1. M denotes the original message. The decrypted data can provide the original information.

In the decryption procedure, the receiver uses their private key. The secret key is subtracted from the standard decryption equation, as shown in Equation (13).

$$M = ((\ C_2 - (P_k * C_1)) - S_k \quad (13)$$

TABLE I. DESCRIPTION OF SENSORS

Sensor	Prototype	Description
Temperature and Humidity sensor	DHT22	It can measure both temperature and humidity levels in the surrounding environment and provides accurate readings through a digital interface.
Temperature sensor	DSB18B20	This is a digital temperature sensor that provides accurate temperature readings with a resolution of up to 12 bits.
UV sensor	GUVA-S12SD	This is a ultraviolet (UV) light sensor that detects UV radiation levels in the surrounding environment. It measures UV intensity across a specific range of wavelengths, typically between 200-400 nm.
Light Intensity sensor	BH1750	This is a digital ambient a 16 bit A-D converter light sensor that measures the intensity of light in its environment.

IV. RESULTS AND DISCUSSIONS

The research presents a robust smart agriculture model implemented in Python for sensor data analysis. Environmental data is collected by sensor nodes, processed, classified via RNN algorithm, and encrypted through hybrid substitution Caesar Cipher and IECC before being transmitted to a cloud server. The cloud server decrypts the data by using their private key and makes decisions based on logical conditions and send messages to farmers. Utilizing real-time sensor data from fields, the system trains a learning model with collected data, categorizing sensor values into four classes: Not required, Short, Medium, and Long. Corresponding class codes are detailed in Table 2.

TABLE II. CODES FOR CLASS

Class	Range (%)	Code
Not Required	0-10	0
Short	10-50	1
Medium	50-75	2
Long	75-100	3

The efficacy of the proposed model is assessed by conducting tests on representative data collected from the sensors. The data collected from various instances is utilized to train and evaluate the proposed data analysis module. The characteristics, including F1-score, accuracy, recall, and precision, are calculated using the data derived from the confusion matrix according to the Equation 14. The acquired values are presented in Table 3 for several categories. Further, False Positive (FP), True Positive (TP), False Negative (FN) and True Negative (TN) values are calculated for the confusion matrix.

$$\begin{cases} F1 = \frac{2TP}{(2TP+FN+FP)} \\ Recall = \frac{TP}{FP+TN} \\ Precision = TP/(TP+FP) \end{cases} \quad (14)$$

TABLE III. PERFORMANCE MATRIX ANALYSIS

Class Code	F1-score	Precision	Recall
0	0.85	1	0.85
1	0.80	0.95	0.98
2	0.95	1	1
3	0.85	0.85	0.95

A. Accuracy

The k-Nearest Neighbor (kNN) based smart monitoring system achieves a maximum accuracy of 65%. In contrast, a comparable smart monitoring system incorporates various classifiers such as Support Vector Machine (SVM), Decision Tree (DT), and Random Forest (RF) to assess the system's accuracy. The achieved accuracies of the aforementioned classifiers are 94.50%, 98.8%, and 98.9%, respectively, which are lower than the accuracy of the suggested model, which is 99.64%. The formula used for calculating the accuracy is given Equation (15).

$$Accuracy = \frac{TN+TP}{TN+TP+FP+FN} \quad (15)$$

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Based on this comparison, it is clear that the suggested model demonstrates better performance than existing alternatives. Figure 5 illustrates a comparative assessment of the accuracy of our proposed method in contrast to various established algorithms.

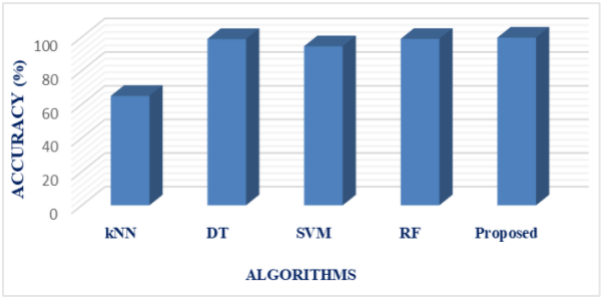


Figure 5. Accuracy comparative analysis

B. *Encryption and Decryption time*

The encryption and decryption times of the proposed hybrid cryptographic algorithm are evaluated and compared with existing cryptographic algorithm such as, DES, 3DES, RSA, Blowfish, and ECC. This comparison is based on the number of connected users, which varies from 5 to 25.

Table IV compares the secure data transfer of the proposed hybrid algorithms and an existing techniques, depending on the time it takes for encryption. Fig. 6 presents a graphical comparison of the average encryption times for sensors data across various existing method. The analysis of the results demonstrates that the suggested hybrid algorithm requires less time for encryption than all other existing algorithm.

TABLE IV. ENCRYPTION TIME COMPARISION (ms)

Numbers of user	DES	3DES	RSA	Blow fish	ECC	Hybrid
5	0.258	0.752	0.333	0.405	0.271	0.133
10	0.503	1.209	0.714	0.751	0.532	0.265
15	0.744	1.932	0.999	0.922	0.812	0.388
20	1.001	2.586	1.336	1.215	1.081	0.541
25	1.238	2.995	1.785	1.345	1.355	0.625
Average	0.749	1.895	1.033	0.928	0.810	0.390

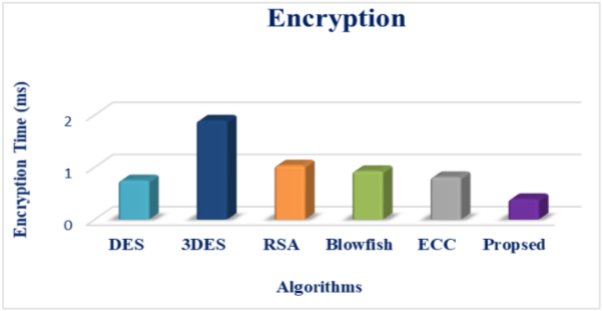


Fig. 6. Average Encryption time

Table V compares the secure data transfer of the proposed hybrid algorithms and an existing techniques, depending on the time it takes for decryption. Fig. 7 presents a graphical comparison of the average decryption times for sensors data across various existing method. The analysis of their results demonstrates that the suggested hybrid method exhibits shorter decryption times compared to other existing methods.

TABLE V. DECRYPTION TIME COMPARIOSION (ms)

Numbers of user	DES	3DES	RSA	Blowfish	ECC	Hybrid
5	0.129	0.385	0.357	0.452	0.493	0.037
10	0.257	0.751	0.762	0.522	0.563	0.07
15	0.384	1.144	1.143	0.622	0.673	0.101
20	0.51	1.42	1.542	0.921	0.988	0.148
25	0.635	1.801	1.785	1.214	1.314	0.195
Average	0.383	1.1002	1.1178	0.7462	0.8062	0.1102

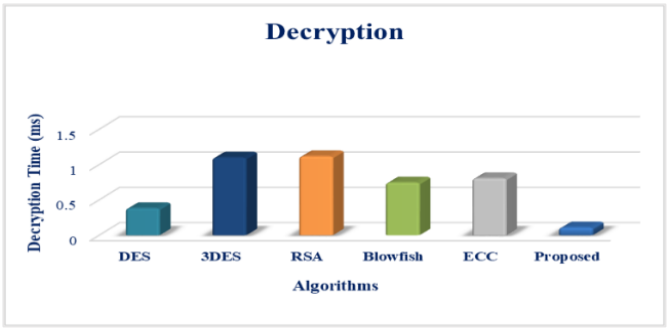


Fig. 7. Average Decryption time

C. *Avalanche Effect*

Equation 16 computes the avalanche effect to measure the differences between two ciphertexts [25]. Fig. 8 illustrates the avalanche effect of the proposed hybrid technique on ten distinct datasets.

$$A = \frac{(\sum ciphertext2 - \sum ciphertext1)}{\sum (ciphertext1)} X100 \quad (16)$$

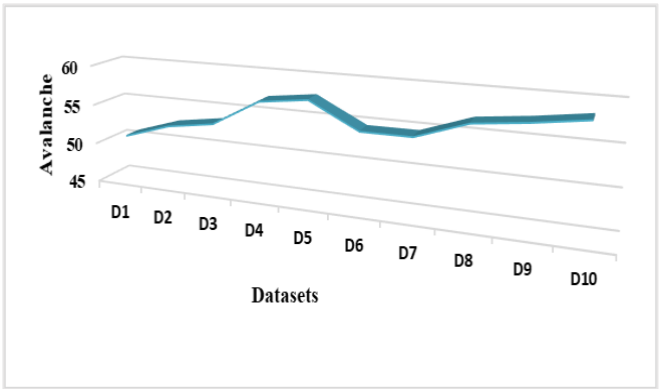


Fig 8. Avalanche effect analysis

V. CONCLUSION

This study introduces a novel approach that implements a robust security measure for IoT-based monitoring systems by integrating RNN with an improved ECC algorithm. The

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RNN enhances data processing for efficient analysis and decision-making from the environmental-based sensor data, to reducing agricultural losses. Additionally, we propose a secure encryption framework combining Substitution Caesar cipher with Improved ECC and an extra secret key for enhancing security. This framework improves encryption and decryption speed and reduces communication overhead. The average encryption and decryption times are 0.3904 ms and 0.1102 ms, respectively. Formal security analysis and simulations demonstrate its effectiveness, with faster encryption and decryption speeds compared to other methods. Statistical analysis confirms its resilience with a significant avalanche effect of 0.55.

REFERENCES

[1] A. Naseer, M. Shmoon, T. Shakeel, S. ur Rehman, A. Ahmad, and V. Gruhn, "A Systematic Literature Review of the IoT in Agriculture - Global Adoption, Innovations, Security Privacy Challenges," IEEE Access, 2024, doi: 10.1109/ACCESS.2024.3394617.

[2] S. Qazi, B. A. Khawaja, and Q. U. Farooq, "IoT-Equipped and AI-Enabled Next Generation Smart Agriculture: A Critical Review, Current Challenges and Future Trends," IEEE Access, vol. 10, pp. 21219–21235, 2022, doi: 10.1109/ACCESS.2022.3152544.

[3] A. N. Lone, S. Mustajab, and M. Alam, " A comprehensive study on cybersecurity challenges and opportunities in the IoT world ," Secur. Priv., vol. 6, no. 6, 2023, doi: 10.1002/spy2.318.

[4] J. Betty Jane and E. N. Ganesh, "Big Data and Internet of Things for Smart Data Analytics Using Machine Learning Techniques," Lect. Notes Data Eng. Commun. Technol., vol. 49, pp. 213–223, 2020, doi: 10.1007/978-3-030-43192-1_24.

[5] R. Manikandan, G. Ranganathan, and V. Bindhu, "Deep Learning Based IoT Module for Smart Farming in Different Environmental Conditions," Wirel. Pers. Commun., vol. 128, no. 3, pp. 1715–1732, 2023, doi: 10.1007/s11277-022-10016-5.

[6] S. Ali and F. Anwer, "Secure IoT framework for authentication and confidentiality using hybrid cryptographic schemes," Int. J. Inf. Technol., 2024, doi: 10.1007/s41870-024-01753-w.

[7] M. S. de Alencar, Cryptography and Network Security. 2022. doi: 10.1515/9781683926900.

[8] L. Jia, X. Zhou, and Q. Wang, "Effects of Agricultural Machinery Operations on PM2.5, PM10 and TSP in Farmland under Different Tillage Patterns," Agric., vol. 13, no. 5, 2023, doi: 10.3390/agriculture13050930.

[9] Kali Charan Rath, Alex Khang, and Debanik Roy, "The Role of Internet of Things (IoT) Technology in Industry 4.0 Economy," Adv. IoT Technol. Appl. Ind. 4.0 Digit. Econ., p. 28, 2024.

[10] A. Morchid, R. El Alami, A. A. Raezah, and Y. Sabbar, "Applications of internet of things (IoT) and sensors technology to increase food security and agricultural Sustainability: Benefits and challenges," Ain Shams Eng. J., vol. 15, no. 3, 2024, doi: 10.1016/j.asej.2023.102509.

[11] A. Aldoseri, K. Al-Khalifa, and A. Hamouda, "A Roadmap for Integrating Automation with Process Optimization for AI-powered Digital Transformation," Preprints, p. 23, 2023, [Online]. Available: https://www.preprints.org/manuscript/202310.1055/v1

[12] "IoT in agriculture: the future of precision monitoring and data-driven farming".

[13] G. Gebresenbet et al., "A concept for application of integrated digital technologies to enhance future smart agricultural systems," Smart Agric. Technol., vol. 5, 2023, doi: 10.1016/j.atech.2023.100255.

[14] R. B. Wakweya, "Challenges and prospects of adopting climate-smart agricultural practices and technologies: Implications for food security," J. Agric. Food Res., vol. 14, 2023, doi: 10.1016/j.jafr.2023.100698.

[15] K. Sanogo, I. Touré, D. D. A. A. Arinloye, E. R. Dossou-Yovo, and J. Bayala, "Factors affecting the adoption of climate-smart agriculture technologies in rice farming systems in Mali, West Africa," Smart Agric. Technol., vol. 5, 2023, doi: 10.1016/j.atech.2023.100283.

[16] Ansari, S. A., & Zafar, A. (2023). Multi video summarization using query based deep optimization algorithm. *International Journal of Machine Learning and Cybernetics*, *14*(10), 3591-3606.

[17] Ansari, S. A., Agrawal, A. P., Wajid, M. A., Wajid, M. S., & Zafar, A. (2024). MetaV: A pioneer in feature augmented meta-learning based vision transformer for medical image classification. *Interdisciplinary Sciences: Computational Life Sciences*, *16*(2), 469-488.

[18] T. Devi, N. Deepa, N. Gayathri, and S. Rakesh Kumar, "AI-Based Weather Forecasting System for Smart Agriculture System Using a Recurrent Neural Networks (RNN) Algorithm," Sustain. Manag. Electron. Waste, pp. 97–112, 2024, doi: 10.1002/9781394166923.ch5.

[19] S. Rani, A. K. Mishra, A. Kataria, S. Mallik, and H. Qin, "Machine learning-based optimal crop selection system in smart agriculture," Sci. Rep., vol. 13, no. 1, 2023, doi: 10.1038/s41598-023-42356-y.

[20] I. H. Sarker, A. I. Khan, Y. B. Abushark, and F. Alsolami, "Internet of Things (IoT) Security Intelligence: A Comprehensive Overview, Machine Learning Solutions and Research Directions," Mob. Networks Appl., vol. 28, no. 1, pp. 296–312, 2023, doi: 10.1007/s11036-022-01937-3.

[21] D. Babuc, "An Authentic Algorithm for Ciphering and Deciphering Called Latin Djokovic," Int. J. Comput. Inf. Eng., pp. 17–10, 2023, [Online]. Available: https://www.researchgate.net/publication/374589454

[22] R. Verma, A. Kumari, A. Anand, and V. S. S. Yadavalli, "Revisiting Shift Cipher Technique for Amplified Data Security," *J. Comput. Cogn. Eng.*, vol. 3, no. 1, pp. 8–14, 2024, doi: 10.47852/bonviewJCCE2202261.

[23] R. Qazi, K. N. Qureshi, F. Bashir, N. U. Islam, S. Iqbal, and A. Arshad, "Security protocol using elliptic curve cryptography algorithm for wireless sensor networks," *J. Ambient Intell. Humaniz. Comput.*, vol. 12, no. 1, pp. 547–566, 2021, doi: 10.1007/s12652-020-02020-z.

[24] N. P. Hegde and P. Deepthi, "Securing Data in Internet of Things (IoT) Using Elliptic Curve Cryptography," *Cogn. Sci. Technol.*, pp. 1013–1020, 2023, doi: 10.1007/978-981-19-8086-2_95.

[25] M. Tahir, M. Sardaraz, Z. Mehmood, and S. Muhammad, "CryptoGA: a cryptosystem based on genetic algorithm for cloud data security," *Cluster Comput.*, vol. 24, no. 2, pp. 739–752, 2021, doi: 10.1007/s10586-020-03157-4.

[26] M. Wasid, S. Ahmad, S. Hussain and M. S. Ansari, "Deep Learning based Object Detection Methods: Current Status and Future Prospects," 2024 7th International Conference on Contemporary Computing and Informatics (IC3I), Greater Noida, India, 2024, pp. 1213-1219, doi: 10.1109/IC3I61595.2024.10829281.

[27] Dinesh Sharma, GS Tomar, "Energy Efficient Multitier Random DEC Routing Protocol for WSN: in Agriculture", Wireless Personal Communication, Vol.120, No.1, pp.727-747, 4th May 2021.

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